CN1 external connector (GM X1, 48-way): OEM vs. new use

OEM: Power Invert Module X1, OEM 98993-5021 / service 19354859. New: open-source replacement logic board, schematic v0.2a.

Pin	OEM use	New use
A1	Not used	Throttle 1 (pedal potentiometer, 0-5 V)
A2	Not used	Throttle 2 (pedal potentiometer, 0-5 V)
A3	Not used	Pedal 5 V supply (50 mA polyfuse)
A4	Not used	Pedal ground
B1	Drive Motor 1 Position Sensor Excitation Signal Negative	Resolver excitation - (same as OEM)
B2	Drive Motor 1 Position Sensor Excitation Signal Positive	Resolver excitation + (same as OEM)
B3	Not used	Pump / fan low-side output
B4	Transmission Range Switch Signal A	Forward input (12 V)
C1	Drive Motor 1 Position Sensor 3 Signal	Resolver SIN - (same as OEM)
C2	Drive Motor 1 Position Sensor 1 Signal	Resolver SIN + (same as OEM)
C3	Transmission Range Switch Signal B	Reverse input (12 V)
C4	Transmission Range Switch Signal P	Brake input (12 V)
D1	Drive Motor 1 Position Sensor 4 Signal	Resolver COS - (same as OEM)
D2	Drive Motor 1 Position Sensor 2 Signal	Resolver COS + (same as OEM)
D3	Transmission Range Switch Signal C	Start input (12 V)
D4	Transmission Range Switch Signal S	Cruise input (12 V)
E1	Not used	Main contactor low-side output
E2	Not used	Precharge contactor low-side output
E3	High Speed GMLAN Serial Data [-] 1	CAN low (same as OEM)
E4	High Speed GMLAN Serial Data [+] 1	CAN high (same as OEM)
F1	Transmission Aux Fluid Pump Relay Control	Not connected (reserved)
F2	Transmission Aux Fluid Pump Feedback Signal	Not connected (reserved)
F3	High Speed GMLAN Serial Data [-] 1	CAN low (same as OEM)
F4	High Speed GMLAN Serial Data [+] 1	CAN high (same as OEM)
G1	High Voltage System Interlock Low Reference 2	Not connected (reserved)
G2	High Voltage System Interlock Signal 3	Not connected (reserved)
G3	High Speed GMLAN Serial Data [-] 3	Not connected (reserved)
G4	High Speed GMLAN Serial Data [+] 3	Not connected (reserved)
H1	Transmission Fluid Temperature Sensor Signal	Not connected (reserved)
H2	Transmission Fluid Temperature Sensor Low Reference	Not connected (reserved)
H3	High Speed GMLAN Serial Data [-] 3	Not connected (reserved)
H4	High Speed GMLAN Serial Data [+] 3	Not connected (reserved)
J1	Traction Power Inverter Motor 1 Temperature Sensor Signal	Motor temperature sensor signal (same as OEM)
J2	Traction Power Inverter Motor 1 Temperature Sensor Low Reference	Motor temperature sensor low reference (same as OEM)
J3	High Speed GMLAN Serial Data [-] 2	Not connected (reserved)
J4	High Speed GMLAN Serial Data [+] 2	Not connected (reserved)
K1	Not used	Encoder A input (encoder assembly option)
K2	Ground	Ground (same as OEM)
K3	Not used	Encoder B input (encoder assembly option)
K4	Not used	Encoder Z input (encoder assembly option)
L1	Internal Mode Switch Voltage Supply A	Not connected (reserved)
L2	Not used	Encoder 5 V supply (200 mA polyfuse)
L3	Run/Crank Ignition Voltage	Run/crank ignition, turns the board on (same as OEM)
L4	Internal Mode Switch Voltage Supply B	Not connected (reserved)
M1	Battery Positive Voltage	Battery + 12 V (same as OEM)
M2	Accessory Wake-Up Serial Data 2	Not connected (reserved)
M3	Not used	Encoder ground
M4	Battery Positive Voltage	Battery + 12 V (same as OEM)